R V UNIVERSITY
SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
B.Sc. (Hons), V SEMESTER
CS3402 – INTERNET OF THINGS LAB
PART B – MINI PROJECT

PHASE 1: PLANNING AND DESIGN (5 MARKS)

Project Title: Smart Room Occupancy Counter

Group No.: 2

Team Members:

S.No	Name	USN
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1. Problem Statement

In classrooms, laboratories, and libraries, manually tracking the number of people present can be inconvenient and inaccurate. This project proposes a low-cost occupancy counter that uses two PIR motion sensors as entry and exit triggers and displays the estimated number of people inside on an LCD. The prototype will be simulated in Tinker-cad.

2. Objective(s)

• Detect motion at the designated entry and exit points using two PIR sensors.
• Increase the occupancy count when the entry sensor is triggered and decrease it when the exit sensor is triggered.
• Display the current count on a 16×2 I2C LCD.
• Prevent the displayed count from dropping below zero.
• Demonstrate the circuit and program in Tinker-cad as a foundation for a future connected monitoring system.

3. Components Required

• Controller: Arduino Uno R3
• Sensors: 2 × PIR motion sensors (entry and exit triggers)
• Output/display: 16×2 I2C LCD
• Supporting components: breadboard, jumper wires, USB power/simulation connection
• Software: Tinker-cad Circuits, Arduino C/C++ code
Note: The current prototype simulates occupancy counting; Wi-Fi remote monitoring is not included in this circuit.

4. Circuit Diagram / Connection Plan

Proposed circuit screenshot is included below. Pin plan:
• Entry PIR: VCC → 5V, GND → GND, OUT → D5
• Exit PIR: VCC → 5V, GND → GND, OUT → D6
• I2C LCD: GND → GND, VCC → 5V, SDA → A4, SCL → A5
• Ensure all components share the Arduino ground. Verify the physical wire endpoints in Tinker-cad before submission.

5. GitHub Repository Link

https://github.com/vishal-h-j/IOT.git

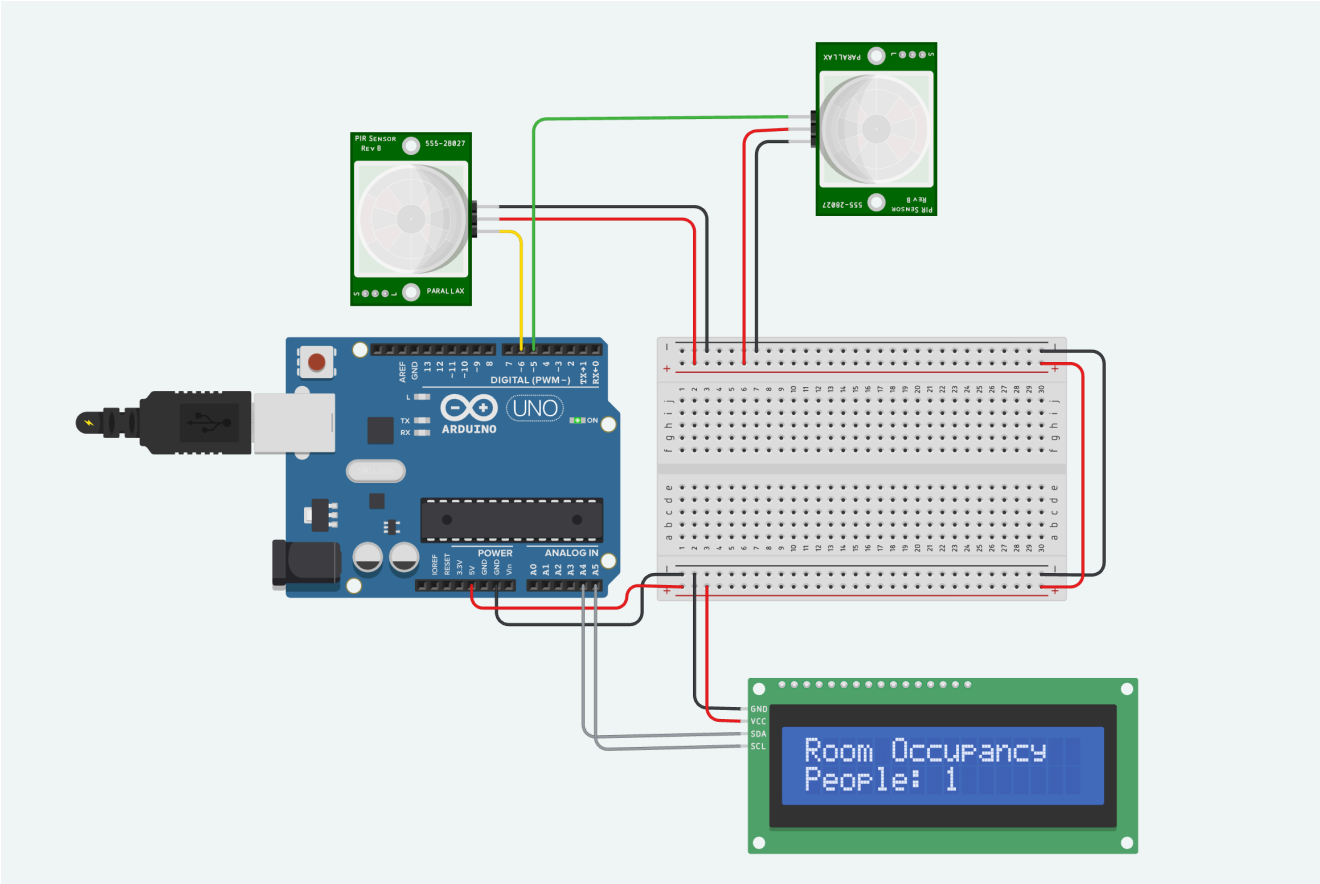


Figure 1. Tinker-cad simulation screenshot showing the Arduino Uno, two PIR sensors, and working I2C LCD test.

Group Review / Faculty Remarks: _____

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Faculty Signature: _____ Marks Awarded: _____ / 5